Control variates for a GARCH model

A Monte Carlo Study with Zero-Variance Estimators

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Outline

Part 1

Introduction

Motivation

The core problem

Bayesian inference on GARCH models requires evaluating intractable posterior expectations, so we rely on **Markov Chain Monte Carlo (MCMC)**. But MCMC estimators can have **high variance**, requiring very long chains to be reliable.

- **Goal:** estimate the posterior mean of $\theta = (\omega_1, \omega_2, \omega_3)$ in a Normal-GARCH(1,1) model efficiently.
- **Key idea:** replace f by a corrected estimator $\tilde{f} = f + \mathbf{a}^T\mathbf{z}$ with the same expectation under π but much smaller variance, using **zero-variance control variates**.
- **Extension:** as the number of control variates grows, we use **LASSO** for dimensionality, and **subsampling** / **block averaging** to validate the regression step.

Based on Mira, Solgi & Imparato, *Stat. Comput.* 23 (2013), 653–662, applied to the DEM/GBP exchange rate data of Ardia (2008).

Part 2

Model and Bayesian Setup

The Model — Normal-GARCH(1,1)

Definition

The return process $\{r_t\}$ follows

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t), \quad h_t = \omega_1 + \omega_2\, r_{t-1}^2 + \omega_3\, h_{t-1}$$

with $\omega_1 > 0,\ \omega_2, \omega_3 \geq 0,\ \omega_2 + \omega_3 < 1$ (covariance stationarity).

- ω_1 — long-run variance floor (background volatility)
- ω_2 — ARCH term: reaction to yesterday's shock
- ω_3 — GARCH term: variance persistence (memory)
- Initialised at $h_0 = \omega_1/(1 - \omega_2 - \omega_3)$

Bayesian Setup

Prior

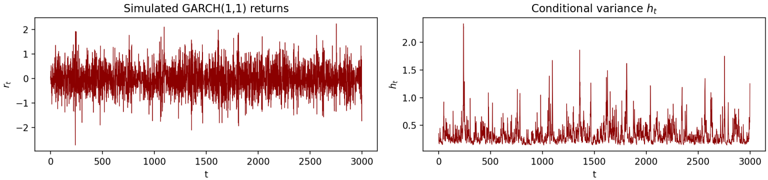
Independent truncated Gaussians: $\omega_j \sim \mathcal{N}(0,1)\,\mathbf{1}_{\omega_j \geq 0},\ j = 1, 2, 3$. Weakly informative — posterior is driven by the data.

Posterior

By Bayes' theorem:

$$\pi(\theta \mid r_{1:T}) \propto \underbrace{\exp\left(-\frac{1}{2}\sum_{t=1}^{T}\left[\log h_t + \frac{r_t^2}{h_t}\right]\right)}_{\text{likelihood}} \times \underbrace{\pi(\theta)}_{\text{prior}}$$

The Simulated Data



T = 3000 returns simulated with θ = (0.055, 0.245, 0.600).

- Returns oscillate around zero (zero-mean by construction) with occasional spikes exceeding $\pm 2.5\%$.
- Visible **volatility clustering** — alternating calm and turbulent stretches, the GARCH signature.
- Conditional variance h_t never reaches zero (floor $\sim$ 0.2 from ω_1); spikes in h_t align with bursts in r_t.
- Unconditional std: $\sqrt{0.055/(1 - 0.245 - 0.600)} \approx 0.6\%$ per day.

Part 3

Sampling the Posterior

The Sampler — Random Walk Metropolis

Why RWM?

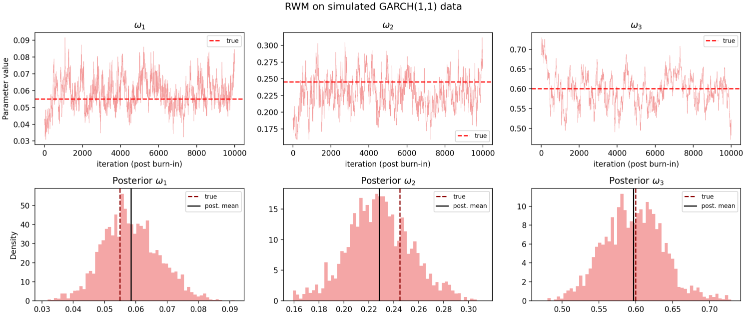
The posterior $\pi(\theta \mid r) \propto L(\theta \mid r)\,\pi(\theta)$ has **no closed form** since the likelihood involves the recursive variance h_t. RWM only needs to *evaluate* π at two points per iteration.

- **Proposal:** $\theta' = \theta + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, \tau^2 I_3)$
- **Acceptance:** $\min\big(1,\ \pi(\theta' \mid r)/\pi(\theta \mid r)\big)$
- Computed in log-scale to avoid underflow
- Positivity enforced via $\log \pi = -\infty$ outside the support

Calibrating τ

Aim for acceptance rate $\approx$ **25%**: too high means the chain barely moves whereas too low means proposals are rejected and the chain freezes.

RWM on Simulated Data

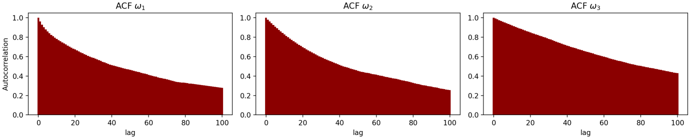


- T = 3000 simulated returns, θ = (0.055, 0.245, 0.600)
- τ = 0.01, acceptance = **27.1%** ✓
- N = 12,000, burn-in = 2,000
- **Identification:** ω_1 & ω_2 partially compensate; ω_3 tightest.

	ω_1	ω_2	ω_3
True	0.055	0.245	0.600
Posterior	0.0585	0.2285	0.5972

Posterior recovery.

The Dependence Problem



Autocorrelation decays slowly: ACF still ≈ 0.4 at lag 100.

Effective Sample Size

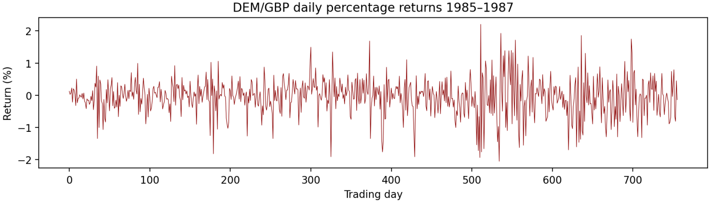
Out of 10,000 post-burn-in draws:

- ESS ω_1 = **70**
- ESS ω_2 = **76**
- ESS ω_3 = **53**

$\sim$ 1 independent sample / 140 iterations.

- Root cause: isotropic proposal does not adapt to the posterior ridge.
- MCMC variance $\gg$ iid case $\Rightarrow$ control variates are needed.

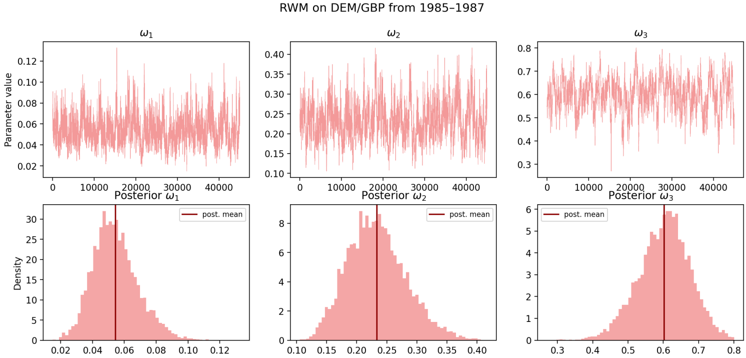
Real Data — DEM/GBP 1985–1987



756 daily returns of the DEM/GBP exchange rate (Ardia, 2008).

- Same dataset used by Mira et al. (2013) for direct comparison.
- Visible volatility clustering: calm stretches alternating with turbulent ones — the empirical motivation for the GARCH model.
- Mean return $\approx 0\%$, std $\approx 0.57\%$ per day.

RWM on Real Data



Trace plots and posterior histograms on 756 daily DEM/GBP returns.

- τ = 0.02, acceptance = **23.1%** ✓
- N = 50,000, burn-in = 5,000
- $\hat{\omega}_2 + \hat{\omega}_3 \approx 0.84$: stationary but persistent volatility

	ω_1	ω_2	ω_3
Ours	0.0545	0.2335	0.6027
Paper	0.055	0.245	0.600

Implementation validated.

PART 4

Variance Reduction: Zero-Variance Control Variates

The Zero-Variance Principle (Mira et al. 2013)

Idea

Replace $f(\theta)$ with a corrected estimator $\tilde{f}(\theta) = f(\theta) + \mathbf{a}^T\mathbf{z}(\theta)$, where $\mathbf{z}$ is **mean-zero** under π. Then $\mathbb{E}_\pi[\tilde{f}] = \mathbb{E}_\pi[f]$: same target but lower variance if $\mathbf{a}$ is chosen well.

Natural choice of z — the score: $\mathbf{z}(\theta) = -\frac{1}{2}\nabla \log \pi(\theta \mid r)$, $\mathbb{E}_\pi[\mathbf{z}] = 0$.

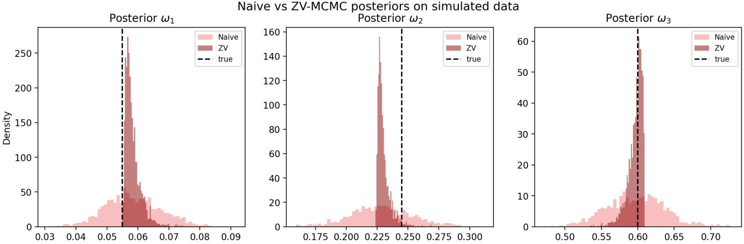
Optimal a — found by **OLS regression** of f on $\mathbf{z}$ along the chain:

$$f(\theta^i) = \alpha + \mathbf{a}^T\mathbf{z}(\theta^i) + \varepsilon_i$$

Intercept $\hat{\alpha}$ is the ZV estimate of $\mathbb{E}_\pi[f]$; residual variance gives $\mathrm{Var}(\tilde{f})$.

For GARCH: $\mathbf{z}$ computed by differentiating the variance recursion $\dot{h}_t^{(j)} = \partial h_t / \partial \omega_j$ via the chain rule.

Zero-Variance Degree-1 — Simulated Data

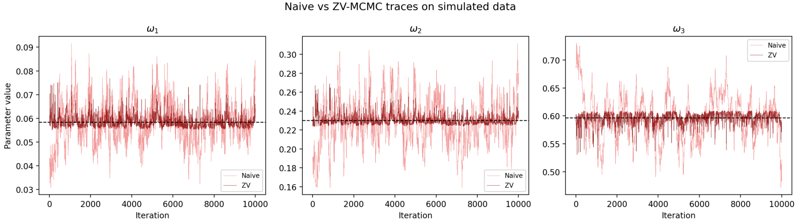


Posteriors visibly tighten: the ZV correction collapses the bulk of the distribution.

	MC std error (naive $\to$ ZV)	Reduction
ω_1	$0.009 \to 0.003$	$12.0\times$
ω_2	$0.025 \to 0.005$	$22.2\times$
ω_3	$0.040 \to 0.010$	$15.4\times$

Std errors in units of 10^{-2}. All within the 95% CI of Mira et al. (2013).

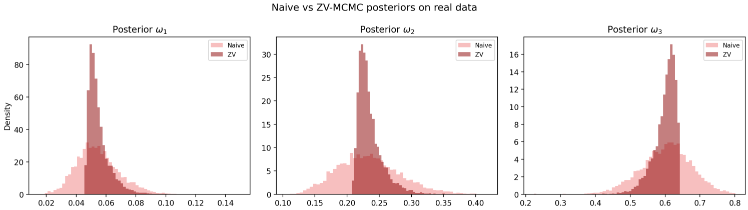
Zero-Variance Degree-1 — Trace Plots (Simulated)



Naive (light) vs. ZV-corrected (dark) chains; dashed line = posterior mean.

- ZV traces oscillate in a **visibly narrower band** around the posterior mean for all three parameters.
- Effect strongest on ω_2 where naive excursions reach 0.16–0.31 versus $\sim$ 0.21–0.26 for ZV.
- The **score-based correction** absorbs most of the chain's high-frequency noise without shifting its centre.
- Same number of iterations, same target — only the per-draw noise is reduced.

Zero-Variance Degree-1 — Real Data (DEM/GBP)

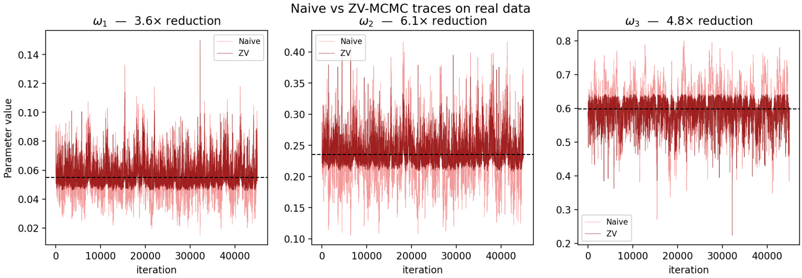


Same pattern, smaller magnitude: the linear $\mathbf{a}^T\mathbf{z}$ explains less of the irregular real-data posterior.

	MC std error (naive → ZV)	Reduction
ω_1	$0.007 \to 0.004$	$3.6\times$
ω_2	$0.023 \to 0.009$	$6.1\times$
ω_3	$0.034 \to 0.016$	$4.8\times$

Real data underperforms $\Rightarrow$ motivates degree-2.

Zero-Variance Degree-1 — Trace Plots (Real Data)



Naive (light) vs. ZV-corrected (dark) chains over 45,000 post-burn-in iterations.

- Variance reduction is **visible but milder** than on simulated data — consistent with the 3.6–6.1× figures.
- Naive chain still shows large excursions (e.g. ω_3 dipping below 0.30) that the ZV chain dampens.
- Linear score correction **cannot fully capture** the curvature of the real-data posterior: residual noise remains.
- Suggests moving to a **quadratic** expansion of the control variate basis.

Scaling Up — Degree-2 Polynomials

The 9 control variates

With $P(\theta) = \mathbf{a}^T\theta + \frac{1}{2}\theta^T B\theta$:

$$\tilde{f}(\theta) = f(\theta) - \frac{1}{2}\mathrm{tr}(B) + (\mathbf{a} + B\theta)^T\mathbf{z}(\theta)$$

- 3 linear: z_1, z_2, z_3
- 6 quadratic: $\omega_j z_k$ for $j \leq k$

Why not just OLS?

Cost $\mathcal{O}(nm^2 + m^3)$, $m = \frac{d(d+3)}{2}$:

Degree	$d = 3$	$d = 10$	$d = 50$
1	3	10	50
2	9	65	1325
3	19	285	23,425

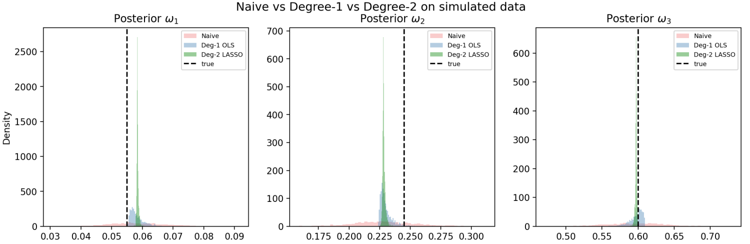
Prohibitive at moderate d.

LASSO (Leluc et al. 2021)

ℓ_1 penalty: $\min_{\mathbf{a}} \|f - H\mathbf{a}\|^2 + \lambda\|\mathbf{a}\|_1$

- Selects relevant variates
- λ tuned by 5-fold CV
- Avoids overfitting on autocorrelated draws
- Robust at large m

Degree-2 LASSO — Simulated Data

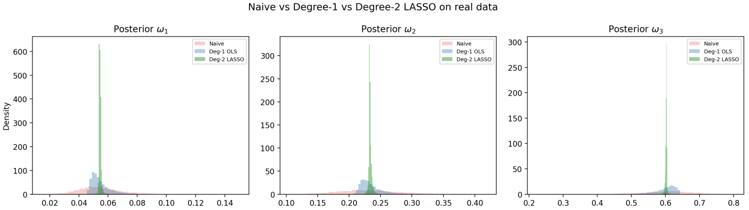


Degree-2 LASSO (green) posteriors collapse onto the true values (dashed) — invisibly thin in the histogram.

Method	ω_1	ω_2	ω_3
Naive	$1\times$	$1\times$	$1\times$
Deg-1 OLS	$12\times$	$22\times$	$15\times$
Deg-2 LASSO	**738**$\times$	**422**$\times$	**1028**$\times$

LASSO selects 9/9 (ω_1, ω_3) and 8/9 (ω_2) variates — no sparsity at $d = 3$.

Degree-2 LASSO — Real Data (DEM/GBP)



Same dramatic collapse on real financial data.

Method	ω_1	ω_2	ω_3
Naive	$1\times$	$1\times$	$1\times$
Deg-1 OLS	$3.6\times$	$6.1\times$	$4.8\times$
Deg-2 LASSO	**298**$\times$	**340**$\times$	**760**$\times$

Three orders of magnitude reduction: it confirms the paper's central finding.

PART 5

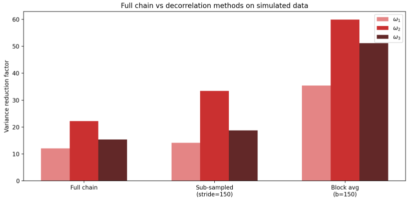
Bonus: Validity of OLS on a Correlated Chain

Bonus — Is OLS Valid on a Correlated Chain?

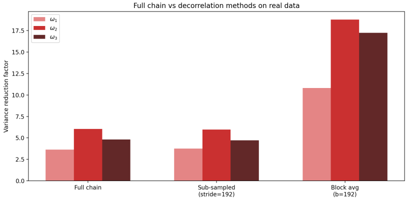
The issue

OLS assumes **i.i.d.** observations. With autocorrelated MCMC draws: $\hat{\boldsymbol{\Sigma}}_{zz}$ uses fewer effective samples than N, coefficients $\hat{\mathbf{a}}$ are less precise, and LASSO cross-validation underestimates uncertainty (adjacent folds nearly identical).

- **Sub-sampling**: keep every n/ESS-th draw
- **Block averaging**: replace each block by its mean



Simulated data — both methods help, block avg triples the reduction.



Real data — sub-sampling barely helps, only block averaging delivers.

Conclusion: block averaging is the robust decorrelation strategy. Moreover, the Q2/Q3 results were *conservative*, not optimistic.

Part 6

Conclusion

Conclusion

- Implemented Bayesian Normal-GARCH(1,1) estimation via RWM with posterior means that do match Mira et al. (2013) within 1%.
- **Degree-1 ZV control variates:** 12–22$\times$ variance reduction (simulated), 4–6$\times$ (real).
- **Degree-2 + LASSO: three orders of magnitude** reduction ($\sim 10^3\times$), confirming the paper's finding and extending it via Leluc et al. (2021).
- Addressed OLS validity on correlated MCMC: block averaging shows our reductions were **conservative**.

Possible extensions

Higher-degree polynomials, adaptive MCMC for better mixing, multivariate or asymmetric GARCH (E-GARCH, GJR-GARCH).

Thank you.

References I

[1] Mira, A., Solgi, R., & Imparato, D. (2013). *Zero variance Markov chain Monte Carlo for Bayesian estimators*. Statistics and Computing, 23(5), 653–662.

[2] Leluc, R., Portier, F., & Segers, J. (2021). *Control variate selection for Monte Carlo integration*. Statistics and Computing, 31(4), 1–26.

[3] Bollerslev, T. (1986). *Generalized autoregressive conditional heteroskedasticity*. Journal of Econometrics, 31(3), 307–327.

[4] Ardia, D. (2008). *Financial Risk Management with Bayesian Estimation of GARCH Models*. Springer.

Appendix — Numerical Error Across Independent Runs